Lung Cancer Detection Using Deep Learning Models: A Comparative Study of Preprocessing Techniques with Explainable AI Integration

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***Abstract*—This work presents an integrated deep learning framework for lung cancer classification from chest CT scans, combining preprocessing optimization, model benchmarking, and explainable AI (XAI). Eleven preprocessing techniques were individually applied to evaluate their impact on classification performance, with Gaussian filtering yielding the best results. A custom convolutional neural network (CNN) enhanced with Convolutional Block Attention Modules (CBAM) was trained alongside four strong baseline architectures, ResNet-50, DenseNet-121, EfficientNet-B0, and ViT-B16, under consistent experimental settings. EfficientNet-B0 with Gaussian filtering achieved the highest performance under the fixed train–validation–test split (97.30%), while cross-validation identified DenseNet-121 with Gaussian filtering as the most consistently reliable configuration. Model interpretability was further assessed using multiple XAI methods, including Grad-CAM, Integrated Gradients, Saliency Maps, Guided Backpropagation, and DeepLIFT, highlighting regions most relevant to clinical decision-making. The findings suggest that the integration of preprocessing optimization, attention-based CNN design, and explainable AI offers a reliable and interpretable approach for lung cancer classification, while future work will aim to validate the framework on larger, multi-institutional datasets.**

***Index Terms*—Lung Cancer, Chest CT Scans, Convolutional Neural Networks, Transfer Learning, Preprocessing Techniques, Explainable Artificial Intelligence (XAI)**

I. Introduction

Lung cancer remains the leading cause of cancer-related mortality worldwide, accounting for nearly 1.8 million deaths in 2022, about 18.7% of all cancer fatalities [1]. Prognosis is highly stage-dependent; early-stage detection greatly improves survival, while late diagnosis severely limits treatment options. Computed Tomography (CT) imaging is the gold standard for early detection and clinical staging, yet its manual interpretation is challenged by subtle radiographic features, reader variability, and the ever-increasing screening workload

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Deep learning, particularly Convolutional Neural Networks (CNNs), has transformed medical image analysis by automatically extracting hierarchical features from raw images, often surpassing traditional methods in classification accuracy [2]. Several studies have explored CNN-based lung cancer detection and subtype classification from CT scans. For instance, Shen *et al.* proposed a multi-crop CNN to distinguish malignant from benign pulmonary nodules with high diagnostic precision [2]. More recently, transfer-learning approaches using pretrained networks such as VGG16 [3], ResNet-50 [4], and EfficientNet [5] have achieved over 97% accuracy on CT datasets [6]. Similarly, Kumaran *et al.* developed an ensemble model combining VGG16 [3], ResNet50 [4], and InceptionV3 and integrated Grad-CAM for interpretability, achieving 98.18% accuracy on lung CT images [7]. Nevertheless, these methods often lack a systematic investigation of how preprocessing choices and interpretability affect diagnostic reliability—two aspects critical for clinical trust and real-world deployment.

Explainable Artificial Intelligence (XAI) techniques have emerged to mitigate the "black-box" limitation of deep neural networks by visualizing and quantifying the image regions that drive predictions [8]. In medical imaging, XAI supports clinical validation by revealing whether model attention aligns with pathological structures, thereby improving transparency and trust. Methods such as Grad-CAM [9], Integrated Gradients [10], Saliency Maps [11], and DeepLIFT [12] have been widely applied across diagnostic imaging tasks to highlight discriminative features or identify potential failure cases. However, despite their growing adoption, comparative evaluations of XAI methods under standardized experimental conditions remain limited in lung cancer analysis, leaving uncertainty about which approaches offer the most reliable and clinically interpretable insights.

In this work, we present a unified and interpretable

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framework for lung cancer classification from chest CT images that systematically integrates preprocessing, model benchmarking, and explainability as illustrated in Fig. 1. We first conduct a comprehensive evaluation of eleven image preprocessing techniques, including CLAHE [13], Gaussian filtering [14], unsharp masking [15], histogram equalization [16], Laplacian enhancement [17], anisotropic diffusion [18], gamma correction [19], texture and multiscale enhancement [20], and morphological operations [21], implemented under consistent training conditions following established medical imaging studies. To strengthen the reliability of the evaluation and address the limitations of the small dataset, the study also incorporates stratified 5-fold cross-validation across model–preprocessing combinations.

II. Methodology

The overall workflow of the proposed lung cancer classification framework is shown in Fig. 1. The process begins with the Kaggle Chest CT Scan Images dataset, followed by the application of eleven preprocessing techniques to enhance image quality and assess their effect on model performance. The preprocessed data are then used to train a custom CNN with CBAM attention and four baseline architectures (ResNet-50, DenseNet-121, EfficientNet-B0, and ViT-B16). Model interpretability is achieved using multiple XAI methods, including Grad-CAM, Integrated Gradients, Saliency Maps, Guided Backpropagation, and DeepLIFT, and final evaluation is performed through accuracy, F1-score, and per-class performance metrics.

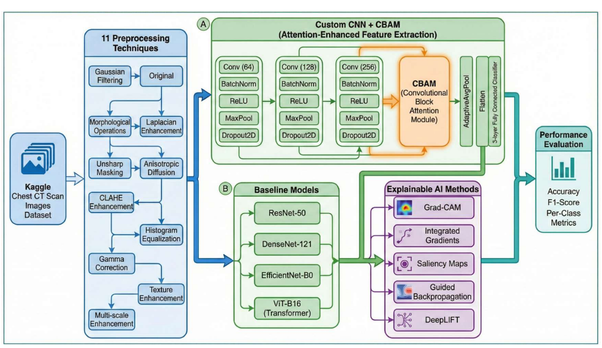

Fig. 1. Deep Learning Architecture and Experimental Pipeline for Lung Cancer CT-Slice Classification With XAI Integration.

A. Dataset

The Chest CT Scan Images dataset [22] is publicly available on Kaggle and specifically designed to support lung cancer detection and classification tasks using supervised machine learning techniques. It contains a total of 1000 chest CT scan images categorized into four diagnostic classes: adenocarcinoma (338 images), large cell carcinoma (187 images), squamous cell carcinoma (260 images), and normal (215 images). To ensure robust model training and evaluation, the dataset was cleaned, resized to 224×224 pixels, and stratified into three subsets: 600 images for training, 200 for validation, and 200 for testing. This stratified splitting preserves the class distribution across all subsets, thereby maintaining class balance during training and evaluation. In addition to this fixed train–validation–test split, the full dataset was further evaluated using a stratified 5-fold cross-validation protocol, as detailed in Section II-D, to obtain statistically robust and leakage-free performance estimates.

B. Preprocessing Techniques

To assess the impact of image enhancement on classification performance, eleven preprocessing techniques were independently applied to the dataset. These included Contrast Limited Adaptive Histogram Equalization (CLAHE) [13], Gaussian filtering [14], unsharp masking [15], morphological operations [21], global histogram equalization, anisotropic diffusion [18], Laplacian enhancement [17], gamma correction [19], texture enhancement, and multi-scale enhancement [20]. Each method was applied to the full dataset prior to training, with models trained and evaluated separately to isolate the effects of each technique on classification accuracy and generalization. All images were resized to 224×224 pixels to ensure consistency across experiments.

C. Model Architecture

The proposed model as shown in Fig.1 is a custom convolutional neural network (CNN) enhanced with Convolutional Block Attention Modules (CBAM) [23] to strengthen spatial and channel-wise feature representation. The network comprises successive convolution–batch normalization–ReLU layers with dropout regularization, followed by adaptive average pooling and fully connected layers for four-class classification. The CBAM units refine feature maps by sequential spatial and channel attention, improving lesion localization in CT scans. The model was trained using the AdamW optimizer [24] with a cosine annealing learning rate schedule to ensure stable convergence. For comparative analysis, four pre-trained architectures, ResNet-50, DenseNet-121, EfficientNet-B0, and ViT-B16, were fine-tuned under the same experimental setup to establish strong performance baselines. In addition to these baselines, three architectures, ResNet-50, CNN+CBAM, and ViT-B16, were included exclusively for the cross-validation experiments to assess the robustness of preprocessing strategies

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across models with varying depth and computational complexity.

D. Training Strategy

To handle class imbalance and improve generalization, several optimization techniques were employed during training. Focal loss [25] was used to down-weight easily classified examples and focus learning on harder cases. Mixup augmentation [26] was applied to combine random image–label pairs, increasing sample diversity and robustness. Label smoothing [27] was further introduced to mitigate overconfident predictions and improve calibration. All models were trained for up to 200 epochs with early stopping, using identical hyperparameters across experiments. Batch normalization was integrated within the network to stabilize learning and enhance convergence efficiency. All models were trained using the same optimization settings and regularization procedures to maintain consistency across experiments.

E. 5-Fold Cross-Validation Protocol

To ensure robust generalization and mitigate the risk of data leakage given the limited dataset size, this study employed stratified 5-fold cross-validation. The dataset was randomly divided into five equally sized, mutually exclusive folds while preserving the original class distribution. For each model-preprocessing combination, four folds were used for training and validation, while the remaining fold served as the test set. This process was repeated five times, with each fold used exactly once as the test set. Final reported metrics represent the aggregated performance across all five folds. This approach provides three key advantages:

- **Mitigation of overfitting:** Models are exposed to diverse training-validation splits
- **Elimination of data leakage:** Class balance is preserved across all folds
- **Reliable comparison:** Enables fair assessment of preprocessing techniques

F. Explainable AI (XAI) Integration

To enhance model transparency and clinical interpretability, several Explainable AI (XAI) techniques were integrated into the evaluation pipeline. Gradient-weighted Class Activation Mapping (Grad-CAM) [9] was employed to generate class-specific heatmaps highlighting regions most influential to the model's predictions. Complementary attribution-based methods, including Integrated Gradients [10], Saliency Maps [11], Guided Backpropagation [28], and DeepLIFT [12] were applied to provide additional perspectives on feature relevance.

III. Results and Discussion

Extensive experiments were conducted to evaluate the classification performance of the proposed CNN framework across 11 different preprocessing techniques as shown in Table I. For each technique, a separate model was trained for 200 epochs with early stopping and evaluated on the test set using accuracy, precision, recall, and F1-score as performance metrics. The results show that Gaussian filtering (Fig. 2) achieved the highest accuracy (93.24%) and F1-score (93.27%), outperforming all other preprocessing methods. Original images and morphological operations followed closely, both yielding an accuracy of 92.57%, while Laplacian enhancement also performed well with an accuracy of 91.89% and an F1-score of 91.95%. Fig. 3 and Fig. 4 show examples of CLAHE enhancement and Histogram Equalization preprocessing techniques for Class 1 (large cell carcinoma).

In contrast, multi-scale enhancement (Fig. 5 exhibited the poorest performance, with an accuracy of 22.3% and an F1-score of 23.70%, indicating that it may introduce noise or irrelevant patterns that hinder effective feature learning. These results demonstrate that preprocessing techniques have a substantial impact on model performance and must be carefully selected in medical image analysis pipelines. This behavior suggests that, rather than enhancing diagnostically relevant structures, the multi-scale transformation may over-amplify intensity variations in a way that disrupts the underlying anatomical patterns required for accurate classification.

To establish a strong performance baseline, four state-of-the-art architectures, ResNet-50 [4], DenseNet-121 [29], EfficientNet-B0 [30], and ViT-B16 [31], were evaluated under the top three preprocessing conditions identified in our earlier experiments (Original, Histogram Equalization, and Gaussian Filtering). The results in Table II show that EfficientNet-B0 with Gaussian Filtering achieved the highest overall accuracy and F1-score (97.30%), followed closely by ResNet-50 with Histogram Equalization (96.64% F1-score) and DenseNet-121 with Gaussian Filtering (96.63% F1-score). Transformer-based ViT-B16 achieved lower accuracy (89.8–93.9%), suggesting that CNN-based architectures remain better suited for limited-data CT analysis. Across all models, Gaussian filtering consistently improved classification, confirming its robustness as a preprocessing method for lung CT images.

Moreover, Table III summarizes cross-architecture accuracy under the three top preprocessing settings. Gaussian filtering provided the most consistent gains for CNNs (EfficientNet-B0: 97.30%, DenseNet-121: 96.62%), whereas histogram equalization favored ResNet-50 (96.62%) and yielded the best results for the transformer baseline (ViT-B16: 93.92%). Original

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images generally underperformed the best enhancement per model. These findings indicate that mild denoising (Gaussian) benefits CNN feature extraction on limited CT data, while transformers appear more sensitive to contrast redistribution (histogram equalization), likely due to their patch/tokenization pipeline.

Furthermore, to strengthen the reliability of the evaluation and mitigate the limitations of the relatively small dataset, all model–preprocessing combinations were further assessed using stratified 5-fold cross-validation. The cross-validated results of the five architectures, ResNet-50, EfficientNet-B0, CNN+CBAM, ViT-B16, and DenseNet-121, across the two preprocessing strategies are summarized in Table IV. Each performance metric represents the mean aggregated score across the five folds, providing a more robust estimate of generalization than a single train/validation/test partition.

Analysis of the cross-validated results reveals several important trends regarding model robustness under different preprocessing strategies. DenseNet-121 consistently achieved the strongest generalization performance, with its best configuration, Gaussian Filtering combined with DenseNet-121, yielding an accuracy of 0.9662 and an F1-score of 0.9661 across the five folds. These results position DenseNet-121 as the most reliable backbone for lung cancer subtype classification in this study. Moreover, Gaussian Filtering provided notable benefits for deeper, high-capacity architectures such as DenseNet-121 and ResNet-50, likely due to its ability to suppress noise and enhance discriminative lesion patterns in CT slices. In contrast, lightweight models such as CNN+CBAM and ViT-B16 exhibited only minor gains or slight reductions with Gaussian Filtering, which may be attributed to their reliance on high-frequency textures that the filtering process attenuates. Overall, the cross-validation analysis reinforces the superiority of the Gaussian Filtering + DenseNet-121 combination, which was therefore selected for subsequent XAI evaluation to confirm that its predictions were driven by clinically meaningful anatomical regions.

In addition to quantitative evaluation, interpretability was assessed using several Explainable AI (XAI) methods: Grad-CAM [9], Integrated Gradients [10], Saliency Maps [11], Guided Backpropagation [28], and DeepLIFT [12]. As shown in Fig. 6, these techniques reveal the model's decision rationale across multiple CT cases. Grad-CAM provided the most spatially coherent and diagnostically relevant activations, consistently focusing on lung regions associated with pathological features in correctly classified samples. In contrast, Integrated Gradients and Saliency Maps produced more diffuse relevance maps, while DeepLIFT often highlighted less interpretable regions. Misclassified examples typically exhibited misaligned or scattered attention patterns across all methods, suggesting uncertainty in the model's internal representation. These findings support the clinical transparency of the proposed system and underscore the utility of XAI in identifying model strengths and potential failure modes.

TABLE I
PERFORMANCE COMPARISON OF PREPROCESSING TECHNIQUES

Technique	Accuracy	Precision	Recall	F1-score
Gaussian Filtering	0.9324	0.9332	0.9324	0.9327
Original	0.9257	0.9272	0.9257	0.9265
Morphological Operations	0.9257	0.9260	0.9257	0.9258
Laplacian Enhancement	0.9189	0.9202	0.9189	0.9195
Unsharp Masking	0.9122	0.9130	0.9122	0.9125
Anisotropic Diffusion	0.8986	0.8991	0.8986	0.8988
CLAHE Enhancement	0.8851	0.8868	0.8851	0.8859
Histogram Equalization	0.8851	0.8862	0.8851	0.8856
Gamma Correction	0.8716	0.8748	0.8716	0.8731
Texture Enhancement	0.8716	0.8827	0.8716	0.8771
Multi-scale Enhancement	0.2230	0.2531	0.2230	0.2370

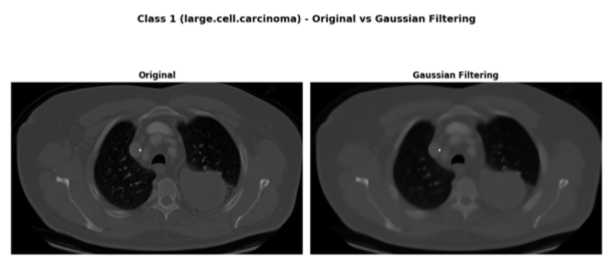

Fig. 2. Gaussian Filtering.

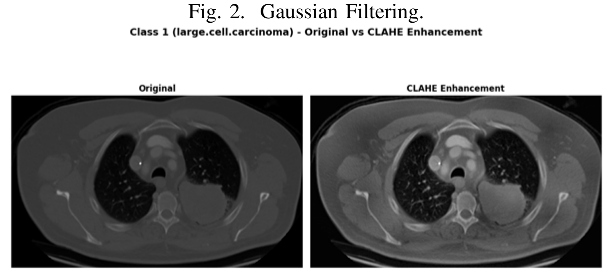

Fig. 3. CLAHE Enhancement.

IV. LIMITATIONS

Despite the promising results, several limitations must be acknowledged. First, the dataset comprised only $1,000$ CT images distributed across four diagnostic categories, which may restrict generalization to diverse patient populations and imaging conditions. However, the use of stratified 5-fold cross-validation helps provide more stable and statistically reliable performance estimates despite the limited sample size. Second, the framework operates on two-dimensional slices rather than volumetric 3D CT data, thereby omitting inter-slice contextual information that could enhance lesion

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TABLE II
BASELINE ARCHITECTURE BENCHMARKING UNDER TOP PREPROCESSING TECHNIQUES

Model + Preprocessing	**Accuracy (%)**	**F1-Score (%)**	**Cohen's kappa (%)**	**MCC (%)**
EfficientNet-B0 + Gaussian Filtering	**97.30**	**97.30**	96.36	96.36
ResNet-50 + Histogram Equalization	96.62	96.64	95.45	95.46
DenseNet-121 + Gaussian Filtering	96.62	96.63	95.44	95.44
EfficientNet-B0 + Original	95.95	95.95	94.53	94.54
DenseNet-121 + Histogram Equalization	95.95	95.96	94.53	94.54
EfficientNet-B0 + Histogram Equalization	95.95	95.94	94.55	94.60
DenseNet-121 + Original	95.95	95.93	94.54	94.54
ResNet-50 + Gaussian Filtering	95.27	95.27	93.62	93.63
ViT-B16 + Histogram Equalization	93.92	93.84	91.79	91.83
ResNet-50 + Original	93.92	94.00	91.81	91.85
ViT-B16 + Original	90.54	90.48	87.23	87.28
ViT-B16 + Gaussian Filtering	89.86	89.78	86.25	86.49

TABLE III
CROSS-ARCHITECTURE ACCURACY BY PREPROCESSING

Preprocessing	**EffNet-B0**	**ResNet-50**	**DenseNet-121**	**ViT-B16**
Gaussian Filtering	**97.30**	95.27	96.62	89.86
Histogram Equalization	95.95	**96.62**	95.95	**93.92**
Original Images	95.95	93.92	95.95	90.54

TABLE IV
5-FOLD CROSS-VALIDATED PERFORMANCE ACROSS MODEL-PREPROCESSING COMBINATIONS

Combination	**Acc**	**Prec**	**Rec**	**F1**
Gaussian Filtering • DenseNet121	**0.9662**	0.9663	0.9662	**0.9661**
Original • DenseNet121	0.9608	0.9607	0.9608	0.9606
Original • ResNet50	0.9567	0.9569	0.9567	0.9567
Gaussian Filtering • ResNet50	0.9486	0.9484	0.9486	0.9483
Original • EfficientNet-B0	0.9459	0.9466	0.9459	0.9459
Original • CNN+CBAM	0.9405	0.9403	0.9405	0.9402
Gaussian Filtering • EfficientNet-B0	0.9391	0.9390	0.9391	0.9389
Original • ViT-B16	0.9310	0.9321	0.9310	0.9309
Gaussian Filtering • ViT-B16	0.9242	0.9252	0.9242	0.9240
Gaussian Filtering • CNN+CBAM	0.9215	0.9214	0.9215	0.9211

characterization. Third, the preprocessing techniques were applied with fixed parameters; more exhaustive hyperparameter tuning could further improve performance. Fourth, the explainability analysis was limited to qualitative methods such as Grad-CAM and Integrated Gradients, as the dataset does not provide lesion- or pixel-level annotations needed for quantitative evaluation of localization accuracy or attribution faithfulness. Finally, model evaluation was limited to retrospective data without external validation. Future work should address these aspects to improve robustness, interpretability, and clinical applicability.

V. CONCLUSION AND FUTURE WORK

This study introduced a unified and interpretable framework for lung cancer classification from chest CT images, integrating a systematic evaluation of preprocessing techniques, benchmarking of deep learning ar-

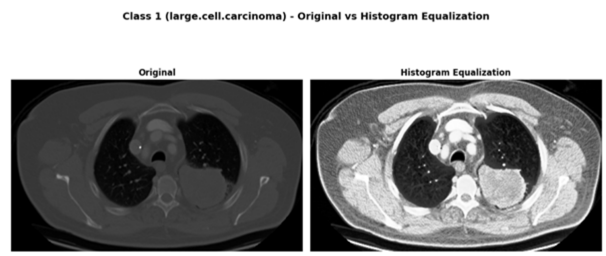

Fig. 4. Histogram Equalization.

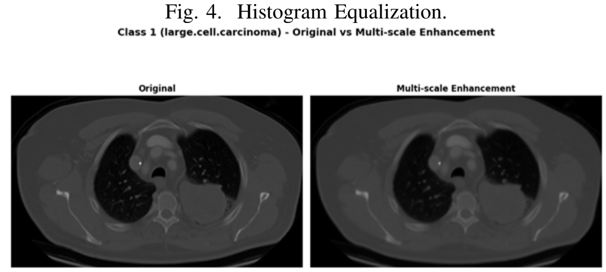

Fig. 5. Multi-scale Enhancement.

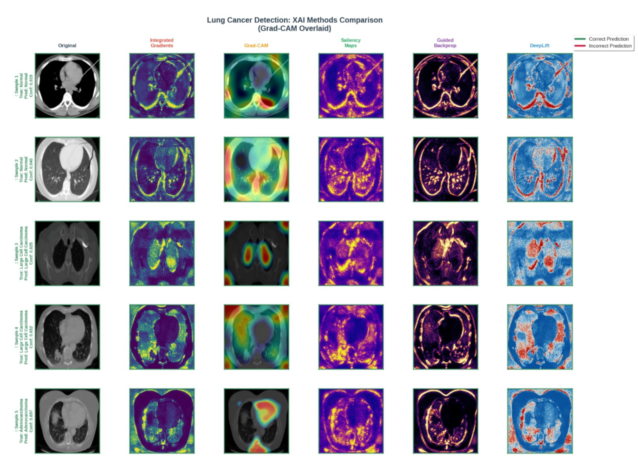

Fig. 6. Comparison of XAI Methods for Lung Cancer Model Interpretability.

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chitectures, and qualitative explainability analysis. Gaussian filtering emerged as the most effective preprocessing method, and while EfficientNet-B0 achieved the highest performance under the fixed train–validation–test split, the stratified 5-fold cross-validation analysis identified DenseNet-121 combined with Gaussian filtering as the most consistently reliable configuration. These findings highlight the substantial influence of preprocessing choices on classification accuracy and demonstrate the value of evaluating multiple architectures under standardized conditions, while Grad-CAM provided the clearest and most clinically aligned visual explanations. Future work will focus on expanding to multi-institutional 3D datasets, exploring transformer-based and hybrid CNN–Transformer architectures, and incorporating quantitative XAI metrics for objective interpretability assessment.

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